Russell Mercer

Designs tools and interfaces for clinicians and researchers

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EDUCATION

Brigham Young University · *Provo, UT* Jan 2024 – Apr 2028 (Expected)
B.S. Biochemistry; Minor in Computer Science, machine learning emphasis
GPA 3.99 · Science GPA 3.99 · MCAT 519 (96th percentile)

DESIGN AND BUILD PROJECTS

Clinical Imaging Workspace · *UT Southwestern* 2026

- Designed and built the interface physicians and medical physicists use: viewer with per-day histograms, parameter comparison, publication image creator, threshold creator, and a fit checker answering whether a number is real
- Iterated from physician feedback; the diffusion overlay let a physician redo 46 contours

Figures, Posters, and Identity · *BYU and UT Southwestern* 2025 – 2026

- Designed a treatment-workflow comparison figure assembled automatically from slide art and R panels, so revisions cost nothing
- Six research posters and talk decks across chemistry, medicine, and structural biology
- Rebuilt the SPRI program mark on an axis model after feedback showed the original read wrong; delivered as an interactive kit

RESEARCH EXPERIENCE

Biochemistry Research Assistant · *Jensen Lab, BYU* Aug 2026 – Present

- Two cryo-electron tomography projects I scoped and the PI approved, awaiting biosafety review; AI agent study presented at the Utah CryoEM Symposium 2026

Clinical Research Assistant · *UT Southwestern Medical Center, Dallas, TX* May 2026 – Present

- Four studies of MR-guided and biology-guided radiotherapy; first author on four manuscripts: one submitted, two in final co-author circulation, one in preparation

Chemistry Research Assistant · *Stowers Lab, BYU* Apr 2024 – May 2026

- Second author, *Langmuir* 2026; built and maintained a flow reactor and a UHV chamber (~10^{-10} Torr, XPS); trained two newer students

LEADERSHIP AND SERVICE

Officer · *BYU AI Association* Sep 2026 – Present

- Lead the weekly AI in Research meeting; speak about AI in research to other colleges across the university and meet with professors and labs

Lead Program Director · *BYU Y-Serve, Early Learning Essentials* Aug 2024 – Present

- Run a volunteer program serving preschoolers from low-income families

Pre-Med Peer Mentor · *BYU Pre-Med Association* Oct 2026 – Present

- Mentor pre-med students; co-taught two university courses as a BYU peer mentor, 2024 – 2025

SKILLS

- **Design:** user research with clinicians, interface design, data visualization, figure and poster design, rapid prototyping
- **Tools:** PowerPoint, ImageJ, ggplot2 and matplotlib, HTML and CSS, SVG
- **Code:** Python, R, C#, Java, AI coding agents
- **Lab and Clinical:** SEM, UHV, cryo-EM sample preparation, clinical data abstraction

PUBLICATIONS

- Ralph AD, **Mercer RW**, et al. *Langmuir* 2026;42(6):4475–88.
- **First author on four manuscripts:** one submitted (*Int J Radiat Oncol Biol Phys*), two in final co-author circulation, one in preparation.
- Co-author on several more in preparation.

AWARDS

- **SCCR Fellow**, BYU Simmons Center for Cancer Research, 2026
- **Outstanding Analytical Chemist**, 2025
- **Dean's List**, four semesters, 2024 – 2025
- **Undergraduate Research Award**, each semester, Stowers Lab

PRESENTATIONS

- Utah CryoEM Symposium, 2026
- BYU Simmons Center for Cancer Research, 2026
- BYU Medical Society, 2026
- ACS Spring Meeting, Atlanta, 2026
- BYU Student Research Conference, 2025 and 2026